Unnumbered continuation: the displacement graph and exact compact-family estimates. Retain all the actual data of C.81 and C.92: the original closed disk $\overline{D}$, its interior O, the compatible metric d on $\overline{D}$, the closed discrete free proper operator group Q, the boundary-fixed centralizer $\mathscr{C}_Q$ with a transitive actual subgroup, and the specified covering $\pi : O \to N = O/Q$. Fix $x \in O$ and $p = \pi x$. Write $\mathscr{F} = \mathscr{C}_Q(x)$. The quotient surface may be noncompact.

Use the exact pointed lifts, image and inverse maps proved in C.92:

$$\begin{aligned} &h_b = \lambda(b), \qquad \pi h_b = b\pi, \qquad h_b(x) = x, \qquad h_b q = q h_b \quad (q \in Q), \\ &\delta_b(\pi z) = d_Q(h_b z, z), \qquad \mathscr{B}_p = \{b \in \mathscr{I}_p : \delta_b \in C_0(N)\}, \\ &D : \mathscr{F} \longrightarrow \mathscr{B}_p, \qquad L : \mathscr{B}_p \longrightarrow \mathscr{F}, \qquad LD = \mathrm{id}_{\mathscr{F}}, \quad DL = \mathrm{id}_{\mathscr{B}_p}. \end{aligned} \tag{CFT.1}$$

Here $\mathscr{I}_p$ is the pointed identity path component in the compact-open topology of maps and inverses, $D(a)(\pi z) = \pi(az)$, and $L(b)$ equals h_b on O and the identity on the original circle. The metric $d_Q(z,w) = \sup_{q\in Q} d(qz, qw)$ retains every original operator. C.81 proves that it gives the original topology on O. C.92 proves that $b \mapsto h_b$ and $b \mapsto h_b^{-1}$ are continuous for compact-open convergence. In particular each lift in this note is the specified lift fixing x.

Put

$$\Delta(b,c) = \sup_{z\in O} d_Q(h_b z, h_c z), \qquad d^{\uparrow}(b,c) = \Delta(b,c) + \Delta(b^{-1}, c^{-1}). \tag{CFT.2}$$

By (C.92.15)–(C.92.16), D and L are isometries for $d^{\uparrow}$ and the original sum of the uniform map and inverse distances on the closed disk. The results below give an exact graph description of this topology and quantitative estimates on its compact families.

An explicit metric for the quotient map-and-inverse topology. Choose the based plane coordinate $\psi : O \to \mathbb{R}^2$ from C.92, and set $K_n = \pi\psi^{-1}(\{v : |v| \le n\})$ for $n \ge 1$. Each K_n is compact, $K_n \subset \operatorname{int} K_{n+1}$, and the interiors cover N. Indeed π is open and the corresponding closed balls lie inside the next open balls. Every compact subset of N lies in some K_n, by a finite subcover of these increasing open sets. Retain the quotient metric $\rho(\pi z, \pi w) = \inf_{q\in Q} d_Q(z, qw)$ from C.81. For pointed homeomorphisms put

$$\begin{aligned} m(b,c) = \sum_{n=1}^{\infty} 2^{-n} \Big[&\min\Big(1, \sup_{y\in K_n} \rho(by, cy)\Big) \\ &+ \min\Big(1, \sup_{y\in K_n} \rho(b^{-1}y, c^{-1}y)\Big) \Big]. \end{aligned} \tag{CFT.3}$$

This is a metric: the triangle inequality follows from that of ρ and $\min(1, a+b) \le \min(1,a) + \min(1,b)$; separation uses $\bigcup K_n = N$. Convergence for m is precisely uniform convergence of maps and inverses on each K_n. In one direction each nonnegative summand tends to zero. In the other, a finite initial sum tends to zero and the remaining sum is bounded by $2\sum_{n>M} 2^{-n}$.

For completeness, uniform convergence on compact sets gives compact-open convergence: for a compact K and an open set V containing $b(K)$, the compact set $b(K)$ has positive ρ-distance from the closed set $N \setminus V$ whenever that complement is nonempty. A sufficiently small uniform change on K therefore stays in V. Conversely fix compact K, a map b, and $\epsilon > 0$. Cover K by finitely many relatively compact neighborhoods U_i whose compact closures satisfy $\rho(bz, by_i) < \epsilon/3$ on $\overline{U}_i$, for some $y_i \in K$. The finite compact-open requirements $c(\overline{U}_i) \subset B_\rho(by_i, 2\epsilon/3)$ imply $\rho(cy, by) < \epsilon$ for every $y \in K$. These neighborhoods exist by continuity of b and local compactness of N. Apply the same arguments to inverses. Thus (CFT.3) gives exactly the stated quotient topology.

The finite-core estimate with the actual lifts. For a nonempty compact $K \subset N$, choose a compact $C \subset O$ with $\pi C = K$. Such a set is obtained from finitely many local covering sections: shrink their domains to cover K by open sets whose closures lie in the section domains, and take

the section images of the corresponding compact pieces of K. Then $\pi^{-1}K = QC$. Equivariance and Q-invariance give the exact equality

$$\sup_{\pi z \in K} d_Q(h_b z, h_c z) = \max_{z \in C} d_Q(h_b z, h_c z). \tag{CFT.4}$$

For a point outside that core, the triangle inequality through the same point z gives $d_Q(h_b z, h_c z) \le \delta_b(\pi z) + \delta_c(\pi z)$. Consequently

$$\Delta(b,c) \le \max\left\{ \max_{z \in C} d_Q(h_b z, h_c z), \quad \sup_{y \notin K} \big(\delta_b(y) + \delta_c(y)\big) \right\}. \tag{CFT.5}$$

An empty tail has supremum zero. The core equality uses the literal equation $h_b(qz) = q h_b(z)$; there is no choice of a different deck representative on any part of this estimate.

Here is the simultaneous inverse control. Let P be a nonempty compact metric space and $u \mapsto b_u$ a continuous family in $\mathscr{B}_p$ for the quotient topology (CFT.3). For a nonempty compact $K \subset N$ define its actual image core

$$K^P = \{ b_u(y) : u \in P,\ y \in K \}. \tag{CFT.6}$$

The joint evaluation map is continuous, so K^P is compact. The exact inverse identity from C.88 is

$$\delta_{b_u^{-1}}(y) = \delta_{b_u}(b_u^{-1} y). \tag{CFT.7}$$

For $y \notin K^P$ one has $b_u^{-1} y \notin K$ for every u. Thus a common bound $\delta_{b_u} < \epsilon$ off K gives the same bound $\delta_{b_u^{-1}} < \epsilon$ off K^P. Choose compact $C^P \subset O$ projecting onto K^P by the same local-section construction. For all $u, v \in P$, (CFT.5) applied to maps and inverses yields

$$\begin{aligned} d^{\uparrow}(b_u, b_v) \le & \max\left\{ \max_{z \in C} d_Q(h_{b_u} z, h_{b_v} z),\ 2\epsilon \right\} \\ & + \max\left\{ \max_{z \in C^P} d_Q(h_{b_u}^{-1} z, h_{b_v}^{-1} z),\ 2\epsilon \right\}. \end{aligned} \tag{CFT.8}$$

The inverse core in this formula is exactly (CFT.6); it does not require a common compact support for the family.

The exact displacement graph, with both inverse maps. Give $C_0(N)$ its supremum-norm topology and define

$$\begin{gathered} \Gamma : \mathscr{B}_p \longrightarrow \mathscr{I}_p \times C_0(N), \qquad \Gamma(b) = (b, \delta_b), \\ \mathscr{G}_p = \{ (b, \delta_b) : b \in \mathscr{B}_p \}, \qquad \Theta : \mathscr{F} \longrightarrow \mathscr{G}_p, \qquad \Theta(a) = \Gamma(D(a)), \\ \Pi : \mathscr{G}_p \longrightarrow \mathscr{F}, \qquad \Pi(b, f) = L(b). \end{gathered} \tag{CFT.9}$$

On the displayed graph f is exactly δ_b. Substitution in (CFT.1) proves $\Pi\Theta = \mathrm{id}_{\mathscr{F}}$ and $\Theta\Pi = \mathrm{id}_{\mathscr{G}_p}$. The inverse of Γ onto its image is the first-coordinate projection.

These are homeomorphisms for the original topology on $\mathscr{F}$ and the product-subspace topology on $\mathscr{G}_p$. Here are both continuity proofs. The triangle inequality for d_Q gives

$$\|\delta_b - \delta_c\|_\infty \le \Delta(b,c), \qquad m(b,c) \le d^{\uparrow}(b,c), \qquad m(b,c) + \|\delta_b - \delta_c\|_\infty \le 2 d^{\uparrow}(b,c). \tag{CFT.10}$$

The middle inequality uses $\rho(b\pi z, c\pi z) \le d_Q(h_b z, h_c z)$ and its inverse counterpart, term by term in (CFT.3), with $\sum 2^{-n} = 1$. Together with the C.92 isometry, (CFT.10) proves continuity of Θ and Γ.

To prove the inverse continuity, take a convergent sequence $(b_n, \delta_{b_n}) \to (b, \delta_b)$ in the graph product. The first coordinates converge for (CFT.3) and the second coordinates converge in

supremum norm. Put $P = \{0\} \cup \{1/n : n \geq 1\}$, with $b_0 = b$ and $b_{1/n} = b_n$. This is a compact continuous quotient family. The set of its displacement functions is compact in $C_0(N)$, because it is a norm-convergent sequence together with its limit. Every compact set of functions in $C_0(N)$ has a common tail: choose finitely many functions within $\epsilon/2$ of every function in the set, choose a compact tail core for each of those finitely many functions with bound $\epsilon/2$, and take their union together with $\{p\}$. Thus $\delta_{b_u} < \epsilon$ outside one nonempty compact K for every $u \in P$.

Use K^P, C, C^P from (CFT.6)–(CFT.8). The joint interior continuity of the lifts from C.92 implies that both core maxima in (CFT.8), with $u = 1/n, v = 0$, tend to zero. To see this directly, the functions $(u, z) \mapsto d_Q(h_{b_u} z, h_b z)$ on $P \times C$ and $(u, z) \mapsto d_Q(h_{b_u}^{-1} z, h_b^{-1} z)$ on $P \times C^P$ are continuous and vanish at every $(0, z)$; a finite cover of each compact core makes the convergence uniform there. It follows that $\limsup_n d^{\uparrow}(b_n, b) \leq 4\epsilon$. As $\epsilon > 0$ is arbitrary, this distance tends to zero. Both the product-subspace topology and $d^{\uparrow}$ are metrizable, so sequential continuity proves continuity. This establishes the claimed homeomorphisms and their specific inverses on the exact graph (CFT.9).

The graph $\mathscr{G}_p$ is closed in $\mathscr{I}_p \times C_0(N)$. Indeed let $(b_n, \delta_{b_n}) \to (b, f)$ in this product. For each $y = \pi z$, compact-open lift continuity and the compatibility of d_Q give

$$\delta_{b_n}(y) = d_Q(h_{b_n} z, z) \longrightarrow d_Q(h_b z, z) = \delta_b(y). \tag{CFT.11}$$

Norm convergence also gives $\delta_{b_n}(y) \to f(y)$. Hence $f = \delta_b \in C_0(N)$, so $b \in \mathscr{B}_p$ and $(b, f) \in \mathscr{G}_p$. Metrizability again makes this sequence argument a proof of closedness.

The continuous common displacement envelope. For the compact quotient family $u \mapsto b_u$ above, define

$$a(y) = \max_{u \in P} \delta_{b_u}(y), \qquad a^-(y) = \max_{u \in P} \delta_{b_u^{-1}}(y). \tag{CFT.12}$$

The functions $(u, y) \mapsto \delta_{b_u}(y)$ and their inverse counterparts are jointly continuous. Their pullbacks by $\mathrm{id}_P \times \pi$ are the continuous functions $d_Q(h_{b_u} z, z)$ and $d_Q(h_{b_u}^{-1} z, z)$, respectively; the product covering is open and hence is a quotient map. Compactness of P then makes both maxima continuous. Explicitly, at y_0 use continuity on product neighborhoods to bound the change in each value by ϵ, take finitely many such parameter neighborhoods covering P, and intersect their y-neighborhoods. This bounds the change of the maximum by the same ϵ.

The following properties of this fixed compact quotient family are equivalent:

1. $u \mapsto L(b_u)$ is continuous in the original closed-disk map-and-inverse topology;
2. $u \mapsto \delta_{b_u}$ is continuous as a map $P \to C_0(N)$ with its supremum norm;
3. $a \in C_0(N)$;
4. the family has the uniform tail in (C.92.17).

The first two are equivalent by the proved graph homeomorphism. Compactness of the image in the second property and the finite-net tail proof above give the third. The third and fourth are the literal equality (CFT.12). For a direct converse from the fourth to the second, choose one compact tail core with bound $\epsilon/3$. On that core, joint continuity and compactness give uniform convergence $\delta_{b_v} \to \delta_{b_u}$ as $v \to u$. Off the core both nonnegative functions are less than $\epsilon/3$, so their difference is less than $\epsilon/3$. Their supremum-norm difference therefore tends to zero. Finally (CFT.6)–(CFT.7) prove that these properties also give $a^- \in C_0(N)$; the same argument proves norm continuity of the inverse displacement family.

For a compact family continuous in the original topology, there is an explicit strictly positive continuous common bound in $C_0(N)$. Construct continuous functions φ_n equal to one on K_n and supported in K_{n+1} as follows. When $\operatorname{int} K_{n+1} \neq N$, put

$$\varphi_n(y) = \frac{\operatorname{dist}_\rho(y, N \setminus \operatorname{int} K_{n+1})}{\operatorname{dist}_\rho(y, N \setminus \operatorname{int} K_{n+1}) + \operatorname{dist}_\rho(y, K_n)}. \tag{CFT.13}$$

The denominator is positive: its two zero sets are the disjoint closed sets $N \setminus \operatorname{int} K_{n+1}$ and K_n. When $\operatorname{int} K_{n+1} = N$, the space N is compact, and take $\varphi_n = 1$. In every case $0 \le \varphi_n \le 1$, $\varphi_n = 1$ on K_n, and its support is compact and contained in K_{n+1}. The uniformly convergent series

$$\tau(y) = \sum_{n=1}^{\infty} 2^{-n} \varphi_n(y), \qquad \eta(y) = a(y) + a^-(y) + \tau(y) \tag{CFT.14}$$

has $\tau > 0$, because every y lies in some K_n. It belongs to $C_0(N)$: outside K_{M+1} the first M summands vanish and the remaining sum is at most $\sum_{n>M} 2^{-n}$. Thus, for each continuous original compact family, (CFT.12)–(CFT.14) define an actual continuous function $\eta \in C_0(N)$ with

$$0 \le \delta_{b_u}(y) \le a(y) < \eta(y), \qquad 0 \le \delta_{b_u^{-1}}(y) \le a^-(y) < \eta(y) \quad (u \in P,\ y \in N). \tag{CFT.15}$$

All maps in this comparison retain the unique pointed lifts and every original deck label. The radial example in (C.92.18)–(C.92.21), with its explicitly assigned Euclidean closed-disk metric, has $b_R \to 1$ for (CFT.3) and $\|\delta_{b_R}\|_\infty > 2/5$. Thus its graph coordinates fail to converge to $(1, 0)$, exactly as its original lifts fail to converge to the identity. This verifies the graph comparison on that previously proved example by the actual coordinate maps (CFT.9).

The proved output here is the closed graph homeomorphism, the finite-core and inverse-core estimates, and the explicit continuous common bound. The noncompact compact-support deformation in C.88 remains unresolved; an actual deformation map has not been constructed in this note. C.91 and C.92, including their stated scopes, remain unchanged.